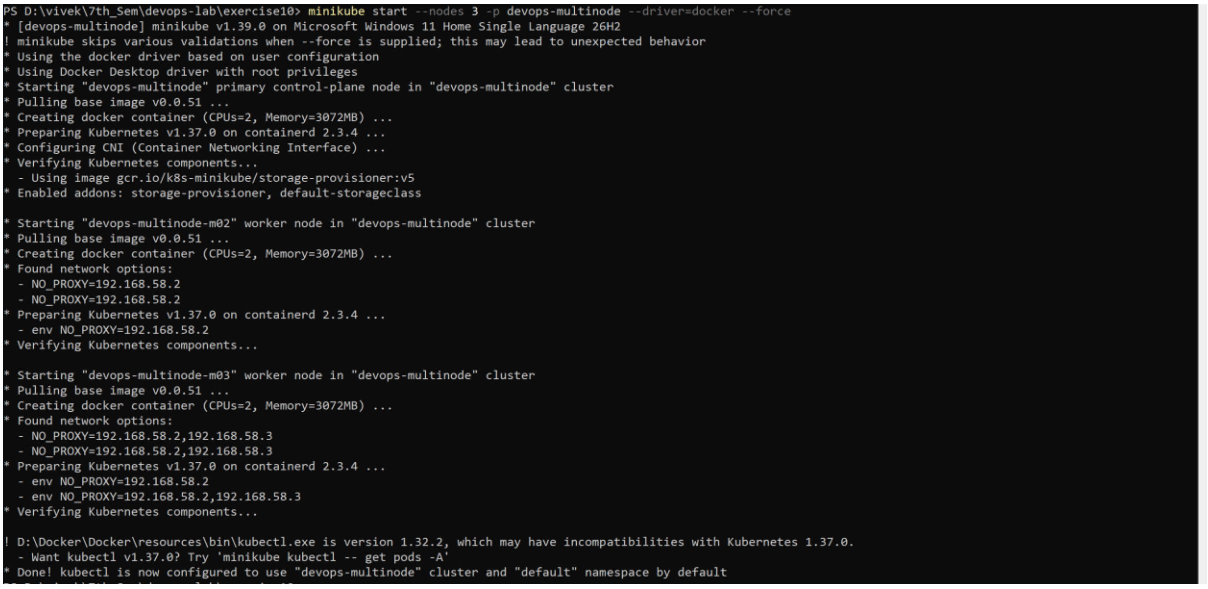
Devops – Exercise 10

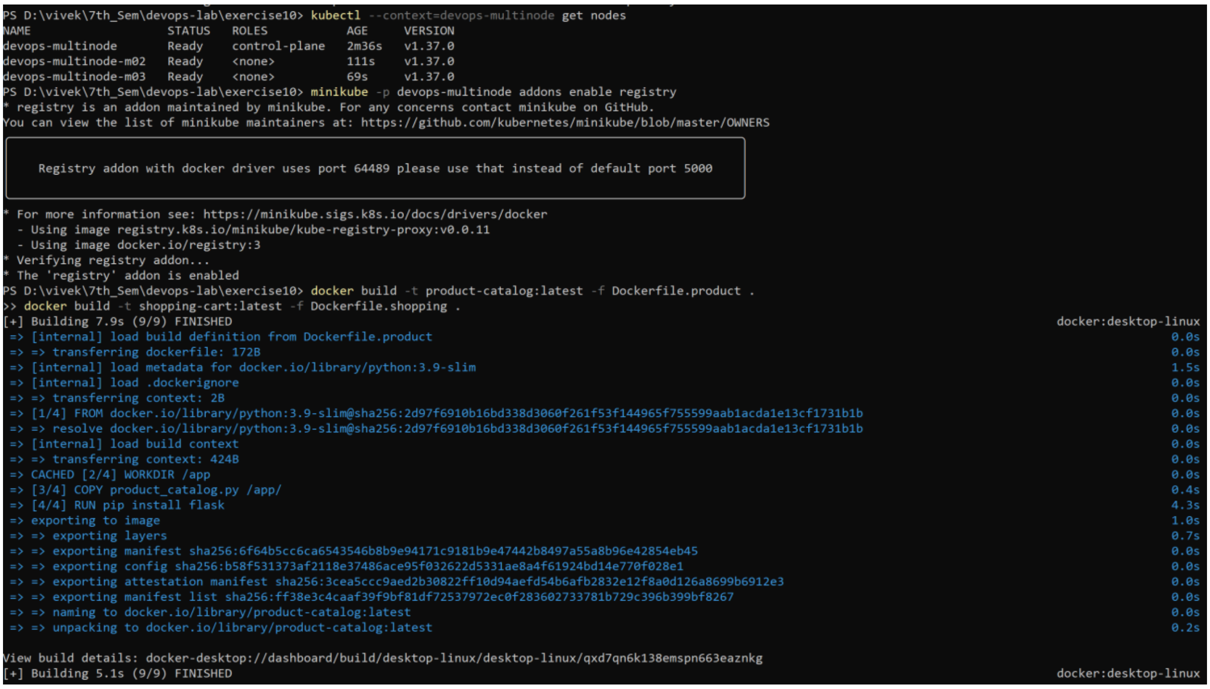


```
PS D:\vivek\7th_Sem\devops-lab\exercise10> minikube start --nodes 3 -p devops-multinode --driver=docker --force
* [devops-multinode] minikube v1.39.0 on Microsoft Windows 11 Home Single Language 26H2
! minikube skips various validations when --force is supplied; this may lead to unexpected behavior
* Using the docker driver based on user configuration
* Using Docker Desktop driver with root privileges
* Starting "devops-multinode" primary control-plane node in "devops-multinode" cluster
* Pulling base image v0.0.51 ...
* Creating docker container (CPUs=2, Memory=3072MB) ...
* Preparing Kubernetes v1.37.0 on containerd 2.3.4 ...
* Configuring CNI (Container Networking Interface) ...
* Verifying Kubernetes components...
  - Using image gcr.io/k8s-minikube/storage-provisioner:v5
* Enabled addons: storage-provisioner, default-storageclass

* Starting "devops-multinode-m02" worker node in "devops-multinode" cluster
* Pulling base image v0.0.51 ...
* Creating docker container (CPUs=2, Memory=3072MB) ...
* Found network options:
  - NO_PROXY=192.168.58.2
  - NO_PROXY=192.168.58.2
* Preparing Kubernetes v1.37.0 on containerd 2.3.4 ...
  - env NO_PROXY=192.168.58.2
* Verifying Kubernetes components...

* Starting "devops-multinode-m03" worker node in "devops-multinode" cluster
* Pulling base image v0.0.51 ...
* Creating docker container (CPUs=2, Memory=3072MB) ...
* Found network options:
  - NO_PROXY=192.168.58.2,192.168.58.3
  - NO_PROXY=192.168.58.2,192.168.58.3
* Preparing Kubernetes v1.37.0 on containerd 2.3.4 ...
  - env NO_PROXY=192.168.58.2
  - env NO_PROXY=192.168.58.2,192.168.58.3
* Verifying Kubernetes components...

! D:\Docker\Docker\resources\bin\kubectl.exe is version 1.32.2, which may have incompatibilities with Kubernetes 1.37.0.
  - Want kubectl v1.37.0? Try 'minikube kubectl -- get pods -A'
* Done! kubectl is now configured to use "devops-multinode" cluster and "default" namespace by default
```



```
PS D:\vivek\7th_Sem\devops-lab\exercise10> kubectl --context=devops-multinode get nodes
NAME                   STATUS   ROLES           AGE     VERSION
devops-multinode       Ready    control-plane   2m36s   v1.37.0
devops-multinode-m02   Ready    <none>          111s    v1.37.0
devops-multinode-m03   Ready    <none>          69s     v1.37.0
PS D:\vivek\7th_Sem\devops-lab\exercise10> minikube -p devops-multinode addons enable registry
* registry is an addon maintained by minikube. For any concerns contact minikube on GitHub.
You can view the list of minikube maintainers at: https://github.com/kubernetes/minikube/blob/master/OWNERS
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│                                                                                                      │
│    Registry addon with docker driver uses port 64489 please use that instead of default port 5000    │
│                                                                                                      │
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* For more information see: https://minikube.sigs.k8s.io/docs/drivers/docker
  - Using image registry.k8s.io/minikube/kube-registry-proxy:v0.0.11
  - Using image docker.io/registry:3
* Verifying registry addon...
* The 'registry' addon is enabled
PS D:\vivek\7th_Sem\devops-lab\exercise10> docker build -t product-catalog:latest -f Dockerfile.product .
>> docker build -t shopping-cart:latest -f Dockerfile.shopping .
[+] Building 7.9s (9/9) FINISHED                                                                                                                   docker:desktop-linux
 => [internal] load build definition from Dockerfile.product                                                                                                       0.0s
 => => transferring dockerfile: 172B                                                                                                                               0.0s
 => [internal] load metadata for docker.io/library/python:3.9-slim                                                                                                 1.5s
 => [internal] load .dockerignore                                                                                                                                  0.0s
 => => transferring context: 2B                                                                                                                                    0.0s
 => [1/4] FROM docker.io/library/python:3.9-slim@sha256:2d97f6910b16bd338d3060f261f53f144965f755599aab1acda1e13cf1731b1b                                           0.0s
 => => resolve docker.io/library/python:3.9-slim@sha256:2d97f6910b16bd338d3060f261f53f144965f755599aab1acda1e13cf1731b1b                                           0.0s
 => [internal] load build context                                                                                                                                  0.0s
 => => transferring context: 424B                                                                                                                                  0.0s
 => CACHED [2/4] WORKDIR /app                                                                                                                                      0.0s
 => [3/4] COPY product_catalog.py /app/                                                                                                                            0.4s
 => [4/4] RUN pip install flask                                                                                                                                    4.3s
 => exporting to image                                                                                                                                             1.0s
 => => exporting layers                                                                                                                                            0.7s
 => => exporting manifest sha256:6f64b5cc6ca6543546b8b9e94171c9181b9e47442b8497a55a8b96e42854eb45                                                                  0.0s
 => => exporting config sha256:b58f531373af2118e37486ace95f032622d5331ae8a4f61924bd14e770f028e1                                                                    0.0s
 => => exporting attestation manifest sha256:3cea5ccc9aed2b30822ff10d94aefd54b6afb2832e12f8a0d126a8699b6912e3                                                      0.0s
 => => exporting manifest list sha256:ff38e3c4caaf39f9bf81df72537972ec0f283602733781b729c396b399bf8267                                                             0.0s
 => => naming to docker.io/library/product-catalog:latest                                                                                                          0.0s
 => => unpacking to docker.io/library/product-catalog:latest                                                                                                       0.2s

View build details: docker-desktop://dashboard/build/desktop-linux/desktop-linux/qxd7qn6k138emspn663eaznkg
[+] Building 5.1s (9/9) FINISHED                                                                                                                   docker:desktop-linux
```
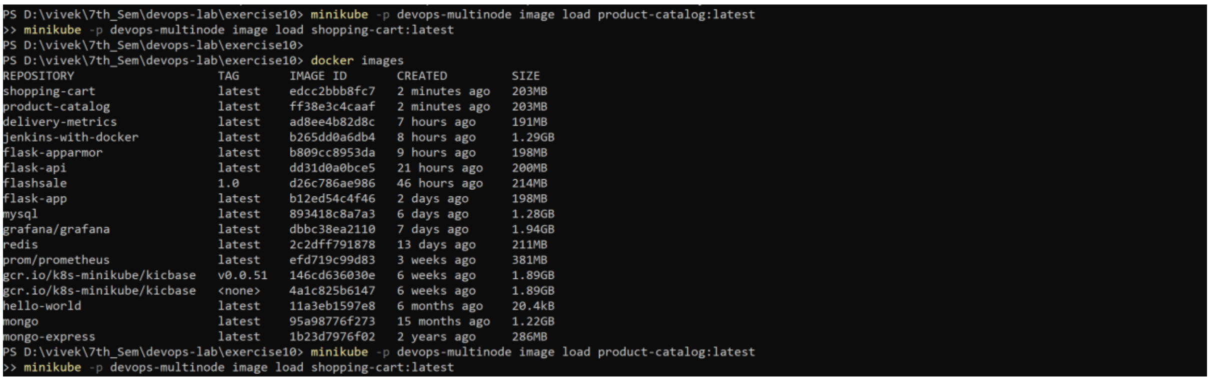
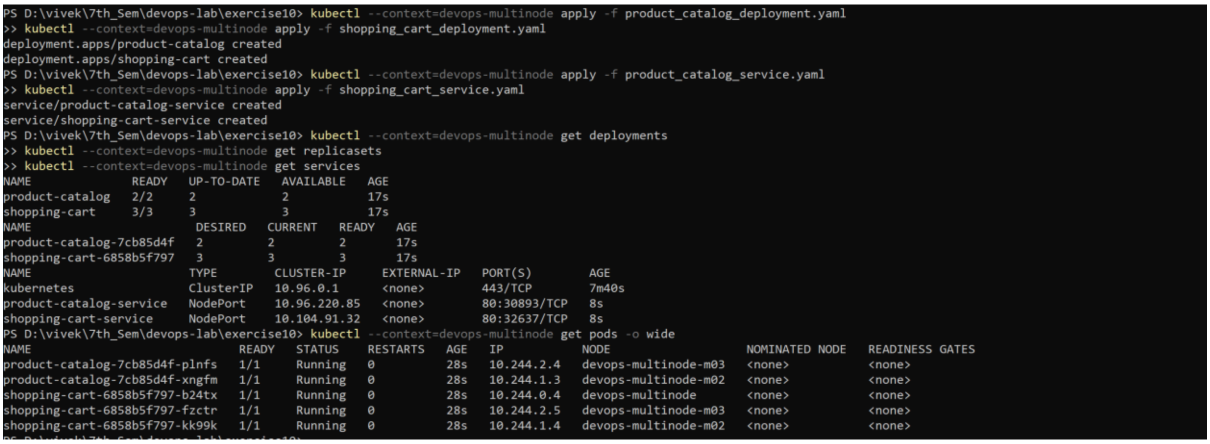
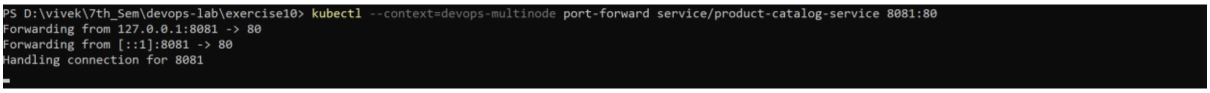



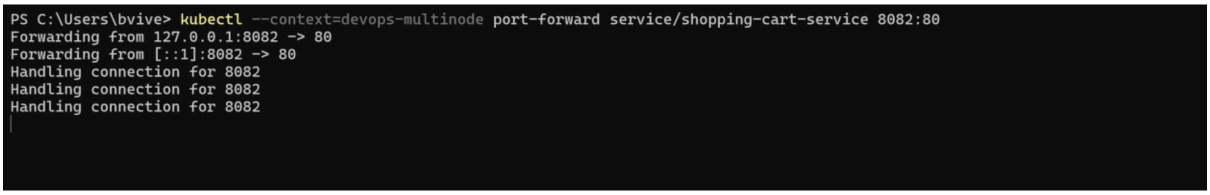
```
PS D:\vivek\7th_Sem\devops-lab\exercise10> minikube -p devops-multinode image load product-catalog:latest
>> minikube -p devops-multinode image load shopping-cart:latest
PS D:\vivek\7th_Sem\devops-lab\exercise10>
PS D:\vivek\7th_Sem\devops-lab\exercise10> docker images
REPOSITORY                    TAG       IMAGE ID       CREATED          SIZE
shopping-cart                 latest    edcc2bbb8fc7   2 minutes ago    203MB
product-catalog               latest    ff38e3c4caaf   2 minutes ago    203MB
delivery-metrics              latest    ad8ee4b82d8c   7 hours ago      191MB
jenkins-with-docker           latest    b265dd0a6db4   8 hours ago      1.29GB
flask-apparmor                latest    b809cc8953da   9 hours ago      198MB
flask-api                     latest    dd31d0a0bce5   21 hours ago     200MB
flashsale                     1.0       d26c786ae986   46 hours ago     214MB
flask-app                     latest    b12ed54c4f46   2 days ago       198MB
mysql                         latest    893418c8a7a3   6 days ago       1.28GB
grafana/grafana               latest    dbbc38ea2110   7 days ago       1.94GB
redis                         latest    2c2dff791878   13 days ago      211MB
prom/prometheus               latest    efd719c99d83   3 weeks ago      381MB
gcr.io/k8s-minikube/kicbase   v0.0.51   146cd636030e   6 weeks ago      1.89GB
gcr.io/k8s-minikube/kicbase   <none>    4a1c825b6147   6 weeks ago      1.89GB
hello-world                   latest    11a3eb1597e8   6 months ago     20.4kB
mongo                         latest    95a98776f273   15 months ago    1.22GB
mongo-express                 latest    1b23d7976f02   2 years ago      286MB
PS D:\vivek\7th_Sem\devops-lab\exercise10> minikube -p devops-multinode image load product-catalog:latest
>> minikube -p devops-multinode image load shopping-cart:latest
```



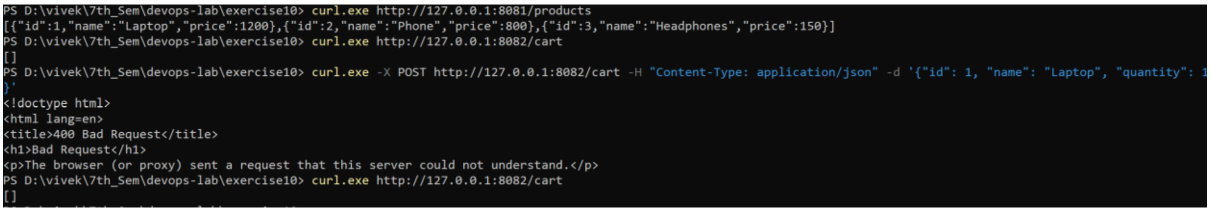
```
PS D:\vivek\7th_Sem\devops-lab\exercise10> kubectl --context=devops-multinode apply -f product_catalog_deployment.yaml
>> kubectl --context=devops-multinode apply -f shopping_cart_deployment.yaml
deployment.apps/product-catalog created
deployment.apps/shopping-cart created
PS D:\vivek\7th_Sem\devops-lab\exercise10> kubectl --context=devops-multinode apply -f product_catalog_service.yaml
>> kubectl --context=devops-multinode apply -f shopping_cart_service.yaml
service/product-catalog-service created
service/shopping-cart-service created
PS D:\vivek\7th_Sem\devops-lab\exercise10> kubectl --context=devops-multinode get deployments
>> kubectl --context=devops-multinode get replicasets
>> kubectl --context=devops-multinode get services
NAME              READY   UP-TO-DATE   AVAILABLE   AGE
product-catalog   2/2     2            2           17s
shopping-cart     3/3     3            3           17s
NAME                         DESIRED   CURRENT   READY   AGE
product-catalog-7cb85d4f     2         2         2       17s
shopping-cart-6858b5f797     3         3         3       17s
NAME                       TYPE        CLUSTER-IP      EXTERNAL-IP   PORT(S)        AGE
kubernetes                 ClusterIP   10.96.0.1       <none>        443/TCP        7m40s
product-catalog-service    NodePort    10.96.220.85    <none>        80:30893/TCP   8s
shopping-cart-service      NodePort    10.104.91.32    <none>        80:32637/TCP   8s
PS D:\vivek\7th_Sem\devops-lab\exercise10> kubectl --context=devops-multinode get pods -o wide
NAME                             READY   STATUS    RESTARTS   AGE   IP            NODE                   NOMINATED NODE   READINESS GATES
product-catalog-7cb85d4f-plnfs   1/1     Running   0          28s   10.244.2.4    devops-multinode-m03   <none>           <none>
product-catalog-7cb85d4f-xngfm   1/1     Running   0          28s   10.244.1.3    devops-multinode-m02   <none>           <none>
shopping-cart-6858b5f797-b24tx   1/1     Running   0          28s   10.244.0.4    devops-multinode       <none>           <none>
shopping-cart-6858b5f797-fzctr   1/1     Running   0          28s   10.244.2.5    devops-multinode-m03   <none>           <none>
shopping-cart-6858b5f797-kk99k   1/1     Running   0          28s   10.244.1.4    devops-multinode-m02   <none>           <none>
```



```
PS D:\vivek\7th_Sem\devops-lab\exercise10> kubectl --context=devops-multinode port-forward service/product-catalog-service 8081:80
Forwarding from 127.0.0.1:8081 -> 80
Forwarding from [::1]:8081 -> 80
Handling connection for 8081
```



```
PS C:\Users\bvive> kubectl --context=devops-multinode port-forward service/shopping-cart-service 8082:80
Forwarding from 127.0.0.1:8082 -> 80
Forwarding from [::1]:8082 -> 80
Handling connection for 8082
Handling connection for 8082
Handling connection for 8082
```



```
PS D:\vivek\7th_Sem\devops-lab\exercise10> curl.exe http://127.0.0.1:8081/products
[{"id":1,"name":"Laptop","price":1200},{"id":2,"name":"Phone","price":800},{"id":3,"name":"Headphones","price":150}]
PS D:\vivek\7th_Sem\devops-lab\exercise10> curl.exe http://127.0.0.1:8082/cart
[]
PS D:\vivek\7th_Sem\devops-lab\exercise10> curl.exe -X POST http://127.0.0.1:8082/cart -H "Content-Type: application/json" -d '{"id": 1, "name": "Laptop", "quantity": 1
}'
<!doctype html>
<html lang=en>
<title>400 Bad Request</title>
<h1>Bad Request</h1>
<p>The browser (or proxy) sent a request that this server could not understand.</p>
PS D:\vivek\7th_Sem\devops-lab\exercise10> curl.exe http://127.0.0.1:8082/cart
[]
```